

Multi-scale Emergent Reality Theory (MER): Formalizing Scale-Dependent Field Topology, Transient Separatrix Dynamics, and the Epistemic Unification of Quantum and Relativistic Regimes

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Abstract

Modern physics remains divided between the discrete, probabilistic paradigm of quantum mechanics (QM) and the continuous, deterministic manifold of general relativity (GR). Multi-scale Emergent Reality (MER) Theory resolves this incompatibility by modeling physical phenomena as scale-dependent projections of a single, continuous universal field governed by conjugate ϕ/ψ expansion-regulation dynamics ($\phi \approx 1.618$, $\psi \approx -0.618$). In this updated framework (v0.2.0), we reformulate the observer scale mismatch parameter ϵ to demonstrate how epistemic quantum uncertainty emerges naturally at subatomic scales ($\epsilon \ll 1$) and collapses into deterministic trajectories via strange attractor basins (e.g., Lorenz-type manifolds) at macroscopic scales ($\epsilon \geq 1$). Furthermore, we update MER's astrophysical and quantum predictions: (1) mapping quantum entanglement directly to a dual-lobe lemniscate separatrix topology ($ER = EPR$), and (2) replacing static modified gravitational potentials with transient field stress decoupling ($T_{\mu\nu}$ tension peaks) during high-energy collisions, providing a pure field-mechanic resolution to galactic cluster dynamics (e.g., the Bullet Cluster) without requiring undiscovered dark matter particles.

1 Introduction

1.1 The Epistemic Gap Between QM and GR

The persistent incompatibility between quantum mechanics and general relativity stems not from a failure of mathematical formulation within their respective domains, but from an unexamined assumption regarding scale neutrality. Standard formulations treat space, time, and probability as intrinsic background properties rather than scale-dependent observations.

MER Theory posits that physical reality is fundamentally deterministic, continuous, and scale-invariant. The apparent non-determinism of the quantum world and the smooth curvature of the cosmological manifold represent two limiting projections of a single field operator observed across varying scale mismatch ratios.

2 Core Mathematical Architecture

2.1 The Conjugate ϕ/ψ Dynamic Operator

We define the universal field evolution operator $\hat{\mathcal{U}}(t)$ as a coupled non-linear system driven by the golden ratio eigenvalues ϕ (expansion vector) and ψ (regulatory/contraction vector):

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.6180339..., \quad \psi = 1 - \phi = -\frac{1}{\phi} \approx -0.6180339... \quad (1)$$

The local field state $\Psi(\mathbf{x}, t)$ evolves under the dual-vector potential:

$$\frac{\partial \Psi}{\partial t} = \phi \nabla^2 \Psi + \psi (\Psi \cdot \nabla) \Psi + \mathcal{S}(\mathbf{x}, \epsilon) \quad (2)$$

where $\mathcal{S}(\mathbf{x}, \epsilon)$ is the scale-dependent interaction term governed by the observer mismatch parameter ϵ .

2.2 Scale Mismatch Parameter ϵ and Epistemic Probability

The observer mismatch parameter ϵ measures the relative spatio-temporal and energetic resolution of the observer (O) relative to the observed system (S):

$$\epsilon = \frac{L_S}{L_O} + \frac{T_S}{T_O} + \frac{E_S}{E_O} \quad (3)$$

- **Quantum Regime** ($\epsilon \ll 1$): The observer's resolution is far coarser than the characteristic action scale of the system. Microscopic phase oscillations fluctuate rapidly within T_O , creating an averaged projection. Probability density P_λ is strictly epistemic:

$$P_\lambda(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \exp \left(-\alpha \frac{\epsilon^2}{\|\phi \Psi + \psi \Psi^*\|^2} \right) \quad (4)$$

- **Classical Regime** ($\epsilon \geq 1$): The system scale exceeds or equals the observer scale. Phase fluctuations average out through self-similar resonance, locking the system into deterministic trajectories.

3 Macro Dynamics and Attractor Filtering

3.1 The Quantum-to-Classical Transition via Strange Attractors

The transition from quantum superposition to classical determinism is governed by non-linear phase-space filtering. At subatomic scales ($\epsilon \ll 1$), unconstrained ϕ/ψ oscillations explore infinite phase pathways. As energy density and scale mismatch increase, multi-scale field feedback acts as a topological filter.

Trajectories are constrained to basins of attraction governed by 3D non-linear manifolds (e.g., the Lorenz attractor):

$$\frac{dx}{dt} = \sigma(y - x) \quad (5)$$

$$\frac{dy}{dt} = x(\rho - z) - y \quad (6)$$

$$\frac{dz}{dt} = xy - \beta z \quad (7)$$

where parameters (σ, ρ, β) are non-linear functions of the ratio ϕ/ψ . Microscopic probabilistic states are not destroyed by wave-function collapse; rather, they are deterministically channelled into the two dominant lobes of the attractor manifold, establishing stable, classical field states.

4 Topological Field Mechanics

4.1 Lemniscate Separatrix and Reconnection Topology

We model localized field concentrations (from subatomic particle pairs to galacto-centric vortices) using a 3D lemniscate (∞) separatrix topology. In polar coordinates (r, θ) :

$$r^2(\theta) = a^2 \cos(2\theta) \cdot \exp \left(\frac{\phi}{\psi} \epsilon \right) \quad (8)$$

This geometry directly mirrors the topology of **magnetic reconnection** in plasma electrodynamics:

1. **The Central X-Point (0, 0):** A spatial null point where opposing vector field lines cross and cancel out. It acts as an energy conversion throat and phase-transition barrier.
2. **The Outer O-Points (Lobes):** Dual, high-tension field islands formed on either side of the X-point during non-linear field stretching.

4.2 Quantum Entanglement as an $ER = EPR$ Lemniscate Throat

Quantum entanglement is reinterpreted without faster-than-light signal propagation. An entangled particle pair consists of the two outer *O*-lobes of a single, continuous lemniscate separatrix, topologically joined at the central *X*-point throat.

When Lobe A is perturbed by measurement:

$$\delta\Psi_A \implies \oint_{\text{separatrix}} \nabla \cdot (\phi\Psi + \psi\Psi^*) dV = 0 \quad (9)$$

Because the separatrix represents a single, closed topological manifold, state resolution at Lobe A instantaneously adjusts the boundary condition at Lobe B. This provides a concrete spatial mechanism aligning MER Theory with the $ER = EPR$ conjecture.

5 Cosmological Applications: The Bullet Cluster & Dark Matter

5.1 Transient Field Stress ($T_{\mu\nu}$) vs. Dark Matter Particles

Prior iterations of MER Theory (v0.1.26) utilized a static logarithmic potential modification $V(r) \approx -\frac{GM}{r}[1 + \phi \ln(r/r_0)]$. While effective for isolated spiral galaxy rotation curves, static potentials fail during high-energy galactic cluster collisions (e.g., 1E 0657-558, the “Bullet Cluster”).

In MER v0.2.0, we eliminate the requirement for physical dark matter particles by modeling galactic collisions as **transient field stress decoupling**:

1. **Baryonic Gas Stalling (Center / X-Point):** During a high-velocity collision, visible intergalactic hydrogen gas interacts via electromagnetic friction and fluid pressure, stalling at the collision zone (*X*-point null region).
2. **Field Lobe Decoupling (Outer O-Points):** The non-linear ϕ/ψ gravitational/field stress tensor $T_{\mu\nu}^{(\phi,\psi)}$ carries the primary space-time curvature forward:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} \left(T_{\mu\nu}^{(\text{baryonic})} + T_{\mu\nu}^{(\phi,\psi) \text{ stress}} \right) \quad (10)$$

3. **Gravitational Lensing Separation:** Gravitational lensing peaks occur at regions of maximum field stress ($T_{\mu\nu}^{(\phi,\psi)}$ tension peaks in the outer *O*-lobes), which naturally detach from the stalled gas core during violent separatrix stretching.
4. **Taylor Relaxation to Equilibrium:** The detached figure-8 field state is temporary. Over cosmic timescales, the outer lobes contract via Taylor relaxation, collapsing back toward a single, balanced toroidal dipole centered on the merged barycenter.

Phenomenon	Standard Model / GR	MER Theory (v0.2.0)
Quantum Uncertainty	Intrinsic fundamental randomness	Epistemic artifact of scale mismatch ($\epsilon \ll 1$)
Entanglement	Spooky action / Wave collapse	Dual O -lobes via central X -point ($ER = EL$)
Galactic Rotations	Cold Dark Matter (WIMPs)	Vacuum field tension (ϕ/ψ stress)
Bullet Cluster	Dark matter pass-through	Transient decoupling of $T_{\mu\nu}^{(\phi,\psi)}$ field lobes

Table 1: Key physical reinterpretations under MER Theory v0.2.0.

6 Summary of Comparative Predictions

7 Conclusion

MER Theory v0.2.0 provides a mathematically consistent, scale-invariant framework unifying quantum mechanics and general relativity. By demonstrating that probability is epistemic, entanglement is topological, and dark matter signatures are transient field stress dynamics, MER offers a clean path forward for unified field research without adding undetected physical particles or background-dependent constants.